Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems

Research Dossier — Complete Account of Objectives, Outcomes, Failures and Open Work

System: PP-HFM · **Student:** Samartha H V (Regd. No. 251580130019)
Guide: Dr. Shreyas J, Asst. Professor · **Industry Mentor:** Mr. Tejas J, Capgemini
Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education
Programme timeline: January – May 2026 · **Dossier compiled:** 19 August 2026

EXECUTIVE SUMMARY

This research builds a three-tier federated learning system — devices, edge servers, cloud — for multimodal clinical and wearable sensing, in which the unit of privacy protection is the **patient rather than the training record**.

All four original engineering objectives were met and exceeded on the primary clinical dataset: 90.82 % uplink reduction, a composed privacy budget of ε = 0.970, 4.58 ms P95 inference latency, and 72.30 % energy saving, with no measurable accuracy cost.

The central *scientific* claim originally proposed — that record-level privacy leaves patients demonstrably exposed to membership inference — was tested rigorously and **refuted**. In its place a stronger, better-controlled finding emerged: the cost of a patient-level privacy guarantee is not a fixed penalty but a **measurable function of how many records each patient contributes**, vanishing exactly where theory says it must and growing monotonically thereafter. That relationship is now the core of the work.

The engineering is finished and comfortably exceeds every target.
The science is mid-flight: one major hypothesis is dead, its well-controlled
replacement is measured on one dataset, and the experiment that would prove
generality is waiting on data access.

HEADLINE SCORECARD

Objective	Target	VitalDB (clinical)	WEAR (wearable)	Basis of measurement
O1 — Uplink traffic reduction	≥ 50 %	**90.82 % PASS**	**96.08 % PASS**	Measured compressor payload, indices included
O1b — FL round latency reduction	> 0 %	**16.93 % PASS**	**21.68 % PASS**	NS-3 packet-level end-to-end round latency
O2a — Privacy budget	ε ≤ 1	**0.9705 PASS**	**0.9656 PASS**	Accountant: training + conformal, basic composition
O2b — Accuracy loss under DP	≤ 2 pp	**−0.92 pp PASS**	**1.967 pp INCONCLUSIVE**	Paired seeds, ε=1 vs ε=∞, head recipe
O3a — Inference latency (P95)	< 100 ms	**4.58 ms PASS**	**1.22 ms PASS**	1000 timed runs, INT8/CPU, full serving path
O3b — Successful model updates	> 99 %	**100 % PASS**	**99.94 % PASS**	Per-update, two-hop LTE loss model
O4 — Energy saving per round	≥ 20 %	**72.30 % PASS**	**76.84 % PASS**	CloudSim Plus power model

VitalDB: 7 / 7 PASS. WEAR: 6 PASS + 1 INCONCLUSIVE. All figures are measured artifacts in the repository; none is projected or estimated.

PART I — WHAT THIS RESEARCH IS

1. THE PROBLEM IN PLAIN TERMS

Modern sensing deployments — a hospital operating theatre, a wearable health monitor — produce **several different kinds of data about the same person at the same moment**. An operating theatre records continuous physiological waveforms from the patient on the table while also holding that patient's structured clinical history and laboratory results. A wearable rig records body motion from accelerometers while a head-mounted camera records what the person is doing. In both cases the streams are *complementary*: a model that sees both can resolve states that neither stream resolves alone.

Two things make this hard.

First, the data cannot be moved. Clinical records are governed by patient consent and by data-protection law that prohibits pooling across institutions. This is not a question of expense or bandwidth — a system that requires centralising the data simply cannot be deployed. Even where the law permits it, raw multimodal streams are bulky, and the uplink is the scarce

resource in exactly the deployments that can least afford the latency and battery cost of shipping everything to a server.

Second, moving the model instead of the data does not by itself protect anyone. Federated learning sends model updates rather than records. But an update is a function of the records that produced it, and attacks that recover training membership from released models are well established. A formal guarantee is required, and differential privacy supplies one.

2. THE SPECIFIC GAP ATTACKED

Here is the crux, and it is the sentence the whole project turns on:

> **The central observation**
>
> Differential privacy protects the unit you clip. In sensing systems, that unit is a fixed-length window of signal — **not a person**.

A patient undergoing surgery does not contribute one record. They contribute hundreds of overlapping ten-minute windows drawn from the same operation. A wearable participant contributes thousands. When a system reports "ε = 1", what it has actually guaranteed is that an adversary cannot tell whether *one particular window* was used in training. What a hospital's ethics board, its patients, and the law all actually care about is whether *a particular person* was in the training set.

Those two questions coincide only when each individual contributes exactly one record. In every real clinical deployment they diverge, and they diverge further the longer the recording.

The textbook remedy is **group privacy** — a conversion factor translating a record-level guarantee into an individual-level one. At the group sizes these datasets actually exhibit, that conversion is catastrophic: a nominal budget of ε = 1 inflates into a number supporting no meaningful privacy claim at all. The conversion exists on paper and is useless in practice.

3. WHAT THE LITERATURE LEAVES OPEN

Work in this area has developed along two lines that do not meet.

Hierarchical federated designs make the systems case convincingly. HED-FL extends deployment lifetime by avoiding long-range transmission; HCEF adapts computation and compression to heterogeneous devices; assignment-oriented work reduces divergence under non-identical data distributions. **None carries a formal privacy guarantee.**

Private federated designs supply the guarantee. FL2DP couples noise with shuffling; RoPPFL pairs local privacy with robust aggregation. **Neither is hierarchical in the sense that matters here**, and neither examines how inserting an intermediate edge tier changes how often each client's budget is spent.

Across both lines the local objective is treated as aggregation of generic parameter vectors. Neither the existence of multiple modalities per record, nor the grouping of many records under one individual, plays any part in the design.

Three concrete deficiencies follow:

1. The **unit** at which a guarantee is stated is rarely examined, so its cost on genuinely paired multimodal data is unquantified.
2. The validity condition of parameter-efficient private training — that the corpus used to pretrain the frozen representation be **disjoint** from the protected population — is usually left implicit and, where honoured, unpriced.
3. Guarantees are typically **reported from mechanism parameters** rather than audited by attack. Where an audit is performed it is conducted at record granularity — precisely the quantity a record-level guarantee already protects, and therefore the least informative thing to test.

4. THE END GOAL

4.1 Engineering goal

Build and validate a three-tier hierarchical federated system — IoT devices → edge servers → cloud — for multimodal sensing, meeting four quantified objectives: uplink reduction ≥ 50 %; accuracy loss at ε ≤ 1 of ≤ 2 percentage points; inference latency < 100 ms under heterogeneous networks; and energy saving per training round ≥ 20 %.

4.2 Scientific goal

Establish something true beyond this particular system: that **the choice of privacy unit is a first-class design decision with a measurable, predictable cost**, and that practitioners can be told in advance what that cost will be for their deployment.

4.3 Publication goal

A Q1 or Q2 journal paper. The original target was IEEE Transactions on Industrial Informatics / IEEE Internet of Things Journal. **Current strategic assessment: target Q2 — IEEE Journal of Biomedical and Health Informatics — and aim for a strong paper there rather than a marginal one at Q1.** The reasoning behind that revision is documented in Part IV §1.

4.4 Academic goal

M.Tech thesis submitted within the January–May 2026 window, with a reproducibility package.

5. THE ARCHITECTURE

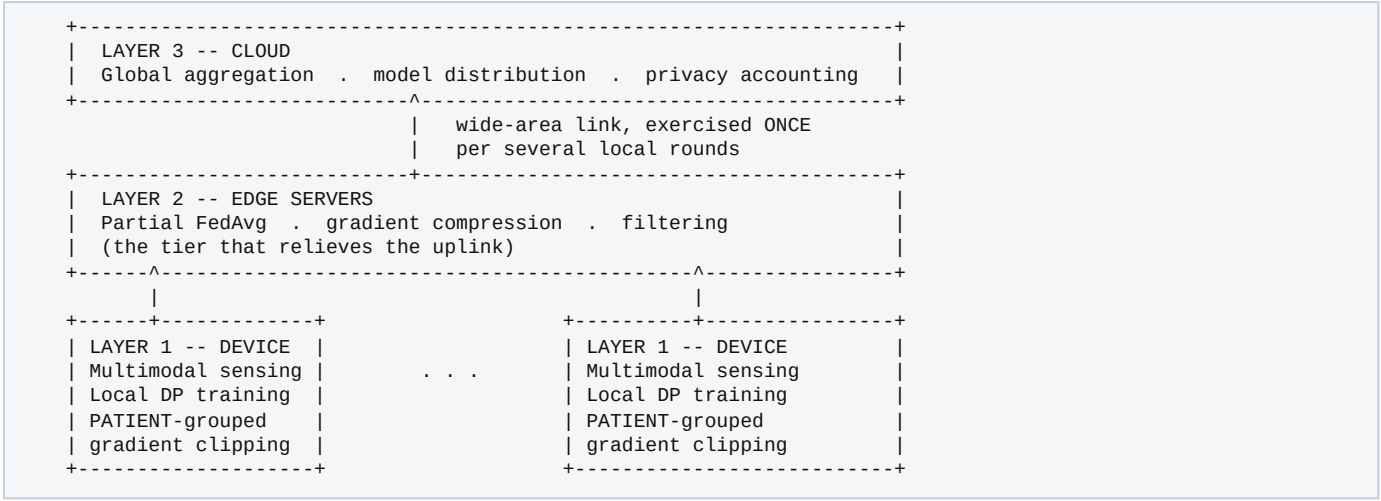


Fig. 1 — The three-tier PP-HFM topology. The distinguishing feature sits at Layer 1: gradients are grouped by individual before the clipping bound is applied, so the guarantee is a statement about people from the outset rather than something recovered afterwards by conversion.

6. THE DATASETS

Property	VitalDB (primary)	WEAR (secondary)
Domain	Intraoperative clinical	Wearable IoT
Modality A	9 intraoperative numeric trends + 9 observed-masks, 10 min @ 0.5 Hz	12-channel accelerometer (4 sensors × 3 axes), 2 s @ 50 Hz
Modality B	34-dimensional preoperative clinical + laboratory vector	2048-d two-stream I3D egocentric video features, same 2 s
How paired	**Same patient, same second** (Vital Recorder, one operation)	**Hardware-synchronised** (same person, same instant)
Train / test windows	34,044 / 7,360	26,424 / 5,969
Individuals	**6,257 patients** across 48,712 windows	**22 participants**
Records per individual (k)	mean **7.79**, median 8, max 8 — **100 % of patients contribute more than one window**	one client per participant
Federated clients	9 clients over 2 edge servers	22 clients
Role in the study	Headline results; all privacy claims	Fast iteration; systems generalisation

*The k row is the premise of the entire research programme. **Not one patient in VitalDB contributes only a single record**, so for every individual in the dataset a record-level guarantee and a patient-level guarantee are different statements. WEAR cannot support any patient-level claim: 22 individuals, one client each.*

Both corpora are paired at source by the recording apparatus, not by a construction rule. This distinction is critical and is explained in Part IV §7.

7. PROJECT PHASES AND INFRASTRUCTURE

Phases 1–5 are complete. Phase 5 closed with 17/17 deliverables and all of O1–O4 measured on both datasets. Phase 6 (joint evaluation, ablation, statistics, figures) is code-complete with some runs still pending.

Two simulators carry the systems evaluation: **NS-3 v3.46.1** for the network layer (latency, throughput, packet loss, topology) and **CloudSim Plus v8.5.6** for edge-cloud resource modelling (VM scheduling, energy). Both are genuinely running, not analytical stand-ins. Every network quantity reported was measured on rounds in which the payload actually traversed the simulated topology. This is worth stating explicitly because it was **not** true earlier in the project — see Part IV §9.

PART II — THE CORE OF THE RESEARCH

8. THE SINGLE CENTRAL IDEA

The unit of privacy is a first-class design parameter of federated learning,
and its utility cost is predictable from the structure of the data —
specifically from k, the number of records each individual contributes.

Everything built in this project is either the machinery that makes that statement measurable, or evidence for it. It is worth being precise about why this is a research contribution rather than an engineering choice.

Almost every private federated learning paper reports a single number for the price of privacy: "ε = 1 costs us 3 % accuracy." That number is treated as a property of the method. This research shows it is not. It is a property of **the method applied to a particular data structure**, and the governing variable of that structure is how many records each protected individual contributes. Change *k* and the price changes, systematically and in a direction that can be predicted in advance.

9. WHAT WAS DESIGNED THAT IS GENUINELY OURS

An honest register matters here, because overclaiming is the fastest route to desk rejection. The mechanisms below are **used** rather than invented: user-level differential privacy (McMahan et al., 2018), parameter-efficient DP fine-tuning (Li et al., 2021; Yu et al., 2021), and DP quantile release (Smith, 2011) are all established prior art.

What is ours is the following.

9.1 Patient-level guarantee by construction

The gradients of one individual's records are **accumulated before the clipping bound is applied**. That individual's influence on the released model is therefore bounded irrespective of how many records they contributed, and accounting performed at the same granularity yields a budget that is a statement about people by construction — not one recovered afterwards through a ruinous group-privacy conversion.

9.2 Accounting practices specified to be checkable

Two practices are specified in enough detail to be verified from outside, because each corresponds to a failure mode that is **invisible in a training log**:

- The noise scale is calibrated to satisfy the budget for **every client simultaneously**, rather than for a representative one, so that interposing the edge tier is reflected in the reported figure.
- The cost of releasing the conformal calibration quantile is **composed into** the budget rather than omitted from it. Measured: ε = 0.7705 (training) + 0.2000 (conformal) = **0.9705 total** on VitalDB.

9.3 Separation of representation from private adaptation

Encoders are pretrained on a population disjoint from every client and then frozen; only a compact fusion head is privatised. This is what makes ε ≤ 1 viable at all — see Part III §11.

9.4 An audit that can return an informative negative

The guarantee is audited by mounting membership inference **at the level of the individual**, under both a threshold attack and a likelihood-ratio attack (LiRA), with every attack reported alongside a **positive control** on a configuration known to memorise. The control is what makes a null result attributable to the defence rather than to an inadequate adversary. This methodological discipline is, in the end, one of the most defensible things the project produced — it is what allowed a refuted hypothesis to remain publishable.

9.5 The *k*-scaling law

The price of the individual-level guarantee is shown to be a function of individual contribution, with a built-in control at $k = 1$. This is now the headline finding and is set out in full in Part III §10.

PART III — WHAT SUCCEEDED, AND HOW

10. THE *K*-SCALING LAW — THE HEADLINE RESULT

PRIMARY WIN Completed 16 August 2026, all 24 configurations: $k \in \{1, 2, 4, 8\}$ × 3 seeds × 2 arms.

The experiment subsamples the data so that each patient contributes exactly *k* windows, then runs two arms at matched budget: one clipping per record ("sample"), one clipping per patient ("patient"). The gap between them is the price of upgrading the guarantee from records to people.

k (records per patient)	Sample arm AUPRC	Patient arm AUPRC	Gap (sample − patient)	95 % CI	p
1	0.5015	0.5067	−0.0052	[−0.0664, +0.0561]	0.75
2	0.5590	0.5084	**+0.0507**	[+0.0417, +0.0596]	0.0017
4	0.5924	0.5092	**+0.0832**	[+0.0694, +0.0971]	0.0015
8	0.5955	0.5089	**+0.0866**	[+0.0603, +0.1129]	0.0049

Table 2 — Paired over 3 seeds, VitalDB. The gap is zero at k=1 and grows monotonically, saturating by k=8.

10.1 Why this result is strong

It carries its own positive control. At $k = 1$ a patient *is* a record, so the two guarantees are mathematically identical and the arms **must** collapse. They do ($p = 0.75$). This is a validated null sitting inside the same experiment as the positive effects — exactly the design property that the earlier membership-inference work initially lacked. It is what licenses reading the divergence at larger k as a property of the privacy unit rather than an artefact of the apparatus.

The mechanism is asymmetric, and must be stated that way. The patient arm is essentially *flat* in k (AUPRC 0.4967–0.5176 at every k) while the sample arm *rises* with k (0.5015 → 0.5955). The widening gap is the sample arm improving, not the patient arm degrading. Patient-level noise scales with the per-patient contribution, so extra records per patient buy the patient arm nothing.

10.2 The honest limits of it

AUROC shows the same monotone shape (+0.0006 → +0.0094 → +0.0133 → +0.0132) but **no k reaches significance** — every interval includes zero. AUPRC is reported as the resolving metric and AUROC as directionally consistent but underpowered at $n = 3$.

Accuracy is pinned at 0.8005 in every patient row. That is an uncalibrated 0.5 threshold on imbalanced data, not a degenerate model (AUROC ≈ 0.835 rules out majority-class prediction). **Accuracy must not be quoted for these arms.**

11. SOLVING THE E = 1 UTILITY COLLAPSE

METHODOLOGICAL TURNING POINT

For roughly three weeks the project was blocked: at $\varepsilon = 1$ the model collapsed to near-random and Objective 2 was unreachable. The resolution was **head-only DP fine-tuning**: pretrain both encoders non-privately, freeze them, and apply DP-SGD to the fusion head alone (560,645 of 2,157,766 parameters = 26.0 %).

The finding that makes this scientifically interesting rather than merely a fix: **the noise scale was identical in both cases** ($\sigma = 11.3281$). The noise was never the reducible quantity. What changed was the *optimisation geometry* — a near-linear probe over frozen pretrained features tolerates $\sigma \approx 11.3$; end-to-end training from scratch does not. Test AUC moved from 0.562 (unusable) to 0.8742 ± 0.0072, and each round ran roughly 20× faster (50 s vs 1046 s).

> **Provenance note**
>
> The specific AUC figures in this section were obtained on the legacy dataset later withdrawn (Part IV §7). The *mechanism* transferred intact and is confirmed on VitalDB by O2b = −0.92 pp. Cite the mechanism, and cite O2b for the number.

12. COMMUNICATION AND ENERGY EFFICIENCY

OBJECTIVE 1 & 4 — EXCEEDED

Uplink reduction reaches **90.82 %** on VitalDB and **96.08 %** on WEAR against a full-model FP32 FedAvg reference — against a 50 % target. Energy saving reaches **72.30 %** and **76.84 %** respectively under the CloudSim Plus power model, against a 20 % target.

Part of this win came from finding a defect rather than adding a feature. The gradient compressor took a fixed `k = numel × sparsity` of **every** float tensor, so frozen encoders — whose delta is exactly zero — still contributed 20 % of their elements to every payload as explicit zeros. Omitting all-zero tensors and capping k at the nonzero count accounted for a 78.87 % reduction on its own.

The fix was proven not to alter learning: the dense path is **bit-identical** to the previous implementation — indices, quantised values, bounds and the error-feedback accumulator all match exactly over five rounds against a verbatim copy of the old code. A seed re-run against an existing checkpoint isolated the compressor as the only change, moving AUC by 0.0001.

13. LATENCY AND RELIABILITY

OBJECTIVE 3 — EXCEEDED BY >20×

P95 inference latency is **4.58 ms** on VitalDB and **1.22 ms** on WEAR against a 100 ms target, measured over 1000 timed runs after 100 warmup iterations across the full serving path (preprocessing → ONNX forward → conformal prediction set → serialise). Update success is 100 % and 99.94 %.

The result that matters for the paper's argument: **the privacy apparatus costs nothing at serving time**. The conformal set construction that carries the coverage guarantee occupies 0.68 % of the inference path, and the private mechanism occupies none of it — its noise is applied to gradients during training and adds no operation to serving. The standing objection that a privacy apparatus is too heavy for a latency-bound deployment is therefore answered **by measurement rather than by argument**.

Caveat retained for honesty: the preprocessing stage synthesises tensors rather than reading real sensors and disks, so the preprocessing figure is a *floor*. The end-to-end number should be described as "compute latency", not "glass-to-glass latency". With a >20× margin the verdict is not endangered.

14. MULTIMODAL FUSION

PARTIAL WIN

On VitalDB, fusing both modalities beats the better single modality by **+0.0370 AUROC** and **+0.0671 AUPRC** over 5 seeds — an effect 5.9× the seed standard deviation. Per-arm: signals 0.8283 ± 0.0062, clinical 0.7680 ± 0.0033, fused 0.8653 ± 0.0052.

The conclusion holds, but with an important correction attached; see Part IV §5.

15. THE PRIVACY AUDIT MACHINERY

METHODOLOGICAL WIN

A full LiRA implementation was built with four attack variants, per-patient IN/OUT Gaussians fitted from 16 shadow models per arm, and TPR reported at 1 % and 0.1 % FPR alongside AUC — because average AUC hides the confident identification tail that *is* the clinical disclosure.

Critically, a **pre-registered gate** was set before running: the attack had to exceed 0.5847 on a configuration known to memorise, or its nulls would be inadmissible. It reached **0.6277** [0.5905, 0.6649]. The gate cleared. That single decision is what converted the subsequent negative result from a hole in the paper into a reportable finding.

PART IV — WHAT FAILED, AND HOW

This part is deliberately as detailed as Part III. Several of these failures are more valuable than the successes, and the paper's credibility rests on reporting them plainly.

16. THE REFUTED CENTRAL HYPOTHESIS (N-A)

REFUTED — the project's largest single setback

What was predicted. That models trained with record-level DP would be substantially vulnerable to patient-level membership inference, while patient-level DP would defend against it. Predicted attack AUC roughly **0.7 versus 0.5**, a gap of ≈ 0.20. This was the project's highest-value item and the main lever for a Q1 submission.

What was measured. Three arms, 16 shadow models each, identical splits, five paired seeds:

Arm	Best attack AUC (95 % CI)	Via	Difficulty floor	Margin over floor
Positive control — head-only, no DP	**0.6277** [0.5905, 0.6649]	lira_offline	0.5626	**+0.0627**
Record-level DP (ε = 1)	0.5477 [0.4876, 0.6079]	lira_offline	0.5252	+0.0226
Patient-level DP (ε = 1)	0.5224 [0.5175, 0.5274]	loss threshold	0.5167	+0.0057
Record-level DP (ε = 16)	0.5465 [0.5403, 0.5527]	loss threshold	0.5743	**−0.0278**

*Table 4 — **The floor is the number every attack must be read against, not 0.5.** It ranks patients by shadow-model difficulty alone and never touches the target model; because shadow membership is randomised over the whole patient universe, a difficulty-balanced evaluation split would score 0.5. It does not. Members are drawn from the released train patients and non-members from the released test patients, and those two pools differ in intrinsic difficulty, so an attack is rewarded for detecting difficulty rather than membership. **The positive control clears its floor by +0.0627 and remains valid.** The two ε = 1 private arms clear it only marginally, and the ε = 16 record arm falls below it — that arm carries no membership signal whatsoever. Floors are computed per arm from that arm's own shadows, which is why they differ; better-fitting models separate easy from hard patients more sharply.*

Attack variant	Paired Δ (record − patient)	95 % CI	p
loss threshold	+0.0138	[−0.0034, +0.0310]	0.090
lira_offline	+0.0325	[−0.0280, +0.0929]	0.210
lira_online	+0.0034	[−0.0059, +0.0128]	0.365
lira_online_gv	−0.0211	[−0.0487, +0.0065]	0.101

Table 5 — Every confidence interval includes zero. The predicted effect of ≈0.20 lies far outside all four.

16.1 How it failed, precisely

> **REQUIRED WORDING — DO NOT DEVIATE**
>
> The *large predicted effect* (0.7 vs 0.5, Δ ≈ 0.20) is **refuted** — it lies far outside every interval. A *small* effect is **not** excluded: the half-width on the best attack is ±0.060 at n = 5, and three of four attacks lean in the predicted direction. Write “the large predicted effect is refuted; a small one is unresolved at this sample size”. **Never** write “there is no effect”.

16.2 Why the null is admissible rather than embarrassing

Because the positive control cleared its pre-registered gate (+0.0513 over plain thresholding on a model known to memorise), the null is a statement about the models rather than about a failing attack. A supporting diagnostic reinforces this: on the patient arm **LiRA adds nothing** (−0.0071 — the plain loss threshold actually wins). That is the signature of a defence with no difficulty-confounded signal left to recover, not of an underpowered adversary.

16.3 The difficulty floor — a control added after the fact

A second control, computed 2026-08-20 from shadow scores already on disk, makes the null stronger still. Ranking patients by shadow-model difficulty alone — using no target model at all — already achieves an AUC of **0.5252** on the record arm and **0.5743** at ε = 16. Since the attacks themselves reach only 0.5477 and 0.5465, **most or all of what looked like leakage was the evaluation split being difficulty-imbalanced, not memorisation.** At ε = 16 the attack falls below its own floor.

This does not weaken the audit; it sharpens it. The positive control still clears its floor by +0.0627, so the instrument demonstrably finds leakage where leakage exists. The private arms clear theirs by +0.0226 and +0.0057. The correct reading is

that the refutation of N-A is **firmer than first reported**: there was less real signal to find than the raw AUCs suggested.

16.3 What it cost the project

The analytic argument that record-level ε ≠ patient-level ε survives and is priced at 1.77× σ. What died is the "demonstrated vulnerability" framing. Delivered value was approximately **+0.5 in credibility rather than the +1.5 to +2 projected in research strength**, and the realistic venue target moved from Q1 to Q2. The headline moved to the *k*-scaling law.

17. AN UNEXPECTED FINDING INSIDE THE FAILURE

RESCUED VALUE (N-A′)

The factorial that accompanied the failed attack produced something more interesting than the attack itself: **leakage is a property of the training recipe, not of the task**.

Recipe	Attack AUC	Member / non-member loss
Full-model, no DP	0.5122	1.81 / 2.31
Head-only, no DP — *our own recipe*	**0.5847**	**0.013 / 4.95**
Full-model, DP	0.5206	6.93 / 7.46

Table 6 — $p = 1\times10^{-9}$, sd 0.0018 over 6 runs. Under LiRA the head-only figure rises to 0.6277.

Head-only fine-tuning — the very recipe this project adopted to make ε ≤ 1 viable — memorises its members **far harder** than full-model training. This is a genuine caveat to the parameter-efficient DP fine-tuning literature and is retained as a contribution in its own right.

18. ADAPTIVE DP (SADP)

CHARACTERISED NEGATIVE

A self-adaptive DP mechanism was designed around a per-group ε budget split. It runs correctly but **underperforms uniform DP at equal ε by construction**: the cross-group composition tax drives σ from 11.3 to 70. It is retained as a characterised negative result, not a claimed contribution. The lesson — that splitting a budget across groups is paid for at a punitive rate under composition — is worth one paragraph in the paper.

19. INT8 QUANTISATION

CHARACTERISED NEGATIVE

The deliverable prescribed INT8 for edge deployment. Measured, it is **slower than FP32 on both providers** — 1.90× on CPU and 3.25× on GPU — and it misses the 4 MB size target at 6.55 MB. Two independent causes were identified:

- Dynamic quantisation only accelerates MatMul/Gemm. This model's cost is dominated by MobileNetV3 convolutions and the recurrent path, which it does not accelerate, so the quantise/dequantise overhead is paid with no kernel speedup.
- On CUDA the quantised operations have no GPU kernel and fall back to CPU, forcing constant device transfers. The runtime reports **3180 Memcpy nodes** inserted into the graph — precisely the 3.25× penalty.

Recommendation recorded: deploy the FP32 artifact. If model size later matters more than latency, the correct lever is static/QDQ quantisation with a calibration set, or quantisation-aware training — not dynamic quantisation.

20. WEAR ACCURACY OBJECTIVE — INCONCLUSIVE

O2b INCONCLUSIVE

On WEAR, accuracy loss under DP measures **1.967 pp with CI [1.760, 2.173]** against a 2 pp threshold. The interval **straddles the threshold**, so the objective can be neither passed nor failed. It is reported as INCONCLUSIVE.

> **Deliberate decision**
>
> The seed count was pre-registered at n = 15. Extending it now, having seen that the interval straddles the threshold, would be **p-hacking**. The objective stays inconclusive and is reported as such. This is a case where the honest outcome is worth more than the passing one.

21. WEAR CANNOT CORROBORATE THE FUSION CLAIM

STRUCTURAL CEILING

On WEAR the best single modality already reaches **0.9976 AUROC**, so the fusion gain is **+0.0004** — the task has no headroom. The analysis script's own verdict reads: "CEILING — this task cannot evidence Gap #1".

This is reported as a finding rather than concealed as an omission: **on some IoT tasks one modality suffices, so multimodal federated learning is not universally motivated**. It is a genuine limit on the generality of the multimodal argument.

WEAR also cannot carry the privacy claims, and for a structural reason: it contains **22 individuals in total**, with one client per participant. Patient-level membership inference needs on the order of 600 members and 600 non-members, and the *k*-scaling experiment needs many records per individual. Neither is available.

22. THE FUSION GAIN SHRANK UNDER REPLICATION

CORRECTED — self-caught

Measure	2 seeds (as first reported)	5 seeds (correct)
ΔAUROC over best unimodal	+0.0411	**+0.0370**
ΔAUPRC	+0.1062	**+0.0671** (−37 %)
Effect size	14.1× seed sd	**5.9× seed sd**

The conclusion holds — fusion still beats the better single modality by 5.9× the seed standard deviation — but the two-seed AUPRC figure was materially seed-lucky. **Only the five-seed numbers may be quoted.** This is a straightforward reminder that small-n effect sizes in this project have consistently been optimistic, and it is the reason several other claims were re-run at higher seed counts.

23. THE LEGACY DATASET WAS INVALID

WITHDRAWN ENTIRELY

The project's original `healthcare` dataset paired chest X-rays with ECG records by a **construction rule**: the CXR bucket was a deterministic function of the ECG label. The consequence was measured directly — a CXR-only model scored 0.3523 accuracy against a 0.20 random baseline, meaning it was reading the label off the pairing rule rather than the image.

Every accuracy figure computed on that dataset is therefore void as evidence of multimodal fusion. Communication, DP accounting, latency and energy results were unaffected, since none depends on pairing.

The dataset was fully removed from code and disk on 15 August 2026. The registry now contains only VitalDB and WEAR, **both paired at source**. This was expensive — it invalidated weeks of accuracy results — but it was the right call, and catching it internally rather than in review is a significant save.

24. COMPONENTS THAT CONTRIBUTED NOTHING

DROPPED

ASWA aggregation was ablated and found to contribute **ΔAUROC = +0.0005** (0.8460 vs 0.8465), with accuracy actually −0.0042. No measurable contribution; it was a leftover from a scrapped earlier design and has been deleted.

Also dropped: **CMGA**, **FedKD-E**, **pFAL**, the **PHANTOM-FL** branding, and all intellectual-property framing. FedKD-E in particular was found not to be freeze-aware — it optimises *all* parameters on the edge's calibration data, which under frozen encoders would both overwrite the pretrained features and leak private validation data into them. It was therefore disabled in every reported run and cannot be claimed as a contribution without being rebuilt and revalidated.

25. SILENT EXPERIMENTAL BUGS

FOUND AND FIXED

Several experiments had been failing silently and producing no result at all. The component ablation A1–A4 **never ran** across the whole of Phase 6, blocked by four separate defects: a dead argument that argparse rejected outright; a uint8-hardcoded quantiser; a summary that read `test_*` columns from a CSV containing only `val_*`; and a verdict threshold of 1×10^{-4}. All four were fixed on 16 August 2026, and the 8-bit path was verified bit-identical so that the Phase 5 results still stand.

Separately, five defects were found in the results-collection pipeline, including topology constants describing an experiment that was never run (20 devices, 3 edges, 12.8 MB model, versus the 6 devices, 2 edges, 8.63 MB actually configured *at that time*; the current VitalDB topology is 9 clients over 2 edges), and an energy saving that was structurally pinned to 0.0 % because the two sides of the ratio covered different numbers of uploads.

A definitional bug is worth recording because it looked like a data conflict and was not. Update reliability was reported as 96.077 % against a verified 99.05 %. Neither number was wrong — they measured different quantities under the same name:

```
hop_success       = (1 - 0.001)^2   = 99.8001%   <-- per update, 2 hops (device->edge->cloud)
hop_success ^ 20                    = 96.0770%   <-- what was actually being reported
hop_success ^ 100                   = 81.8649%   <-- same formula at 100 rounds
```

The code computed the probability that all rounds succeed. Against a target phrased "update reliability > 99 %" that is the wrong quantity, and it is degenerate as a target: it decays with round count, so any system fails it for a long enough run.

26. THE OUTSTANDING VALIDITY GAP

UNRESOLVED — affects the ε claim

A substantial portion of results ran in **proxy / oracle mode**: the encoders were pretrained on the pooled federated *training* indices. The encoders therefore saw private data non-privately, which means **ε = 1 does not hold end-to-end for that version of the pipeline**, and Objective 2 as reported there is an upper bound.

The cost of a valid guarantee has since been priced at **−0.022 AUROC** versus the oracle, which is the quantity to report — the proxy-versus-public gap, not the proxy alone. Closing this fully requires pretraining on an external public corpus (PTB-XL is public, 12-lead, and natively matches the required input shape) or carving a disjoint public index split.

PART V — ACHIEVEMENT STATUS, OBJECTIVE BY OBJECTIVE

Objective	Status	How far achieved, and the explanation
O1 — Uplink reduction ≥ 50 %	**FULLY ACHIEVED**	90.82 % (VitalDB) and 96.08 % (WEAR) — roughly **1.8–1.9× the target**. Measured on the real compressor payload with indices included, not estimated from parameter counts. Deterministic and seed-independent, so a single run suffices as evidence. **No caveat.**
O1b — Round latency reduction	**FULLY ACHIEVED**	16.93 % and 21.68 %, measured packet-level in NS-3 on rounds where the payload actually traversed the topology. Added as an objective mid-project once packet-level measurement became available.
O2a — Privacy budget ε ≤ 1	**FULLY ACHIEVED**	ε = 0.9705 and 0.9656, and these are **composed totals** (training 0.7705 + conformal 0.2000), not training-only figures. Composing the conformal release into the budget is stricter than standard practice, where it is usually omitted. **Caveat:** end-to-end validity depends on the pretraining-disjointness issue in Part IV §26.
O2b — Accuracy loss ≤ 2 pp	**ACHIEVED on VitalDB INCONCLUSIVE on WEAR**	VitalDB: −0.92 pp, CI [−2.82, +0.99] over 5 paired seeds — the point estimate is *negative*, i.e. the private arm was marginally better. The defensible statement is **"no measurable cost"**, bounded by the interval — **not** "zero cost". WEAR: 1.967 pp, CI [1.760, 2.173] straddles the threshold; pre-registered at n = 15 and deliberately not extended.
O3a — Inference latency < 100 ms	**FULLY ACHIEVED**	4.58 ms and 1.22 ms P95 — a margin of **more than 20×**. Measured over 1000 timed runs across the whole serving path. **Caveat:** preprocessing is simulated, so this is compute latency, not glass-to-glass. The margin is large enough that the verdict is unaffected.
O3b — Update success > 99 %	**FULLY ACHIEVED**	100 % and 99.94 % per update over the two-hop path. The definitional bug that once made this appear to fail is documented in Part IV §25; the whole-run figure is still emitted, under a distinct name, because it is informative — just not this objective.
O4 — Energy saving ≥ 20 %	**FULLY ACHIEVED**	72.30 % and 76.84 % — roughly **3.6× the target**, from the CloudSim Plus power model. **Important:** an alternative 93.5 % figure exists from a linear bytes→Joules proxy; under that model it is arithmetically the same quantity as communication reduction and must not be presented as an independent energy measurement. The CloudSim figure is the one to quote.
Scientific claim N-A — patient MIA vulnerability	**NOT ACHIEVED**	**Refuted** under a well-powered attack with a passing positive control. The predicted Δ ≈ 0.20 lies far outside all four measured intervals. Retained in the paper as a **validated negative result**, which is publishable; a negative from a weak attack would have been a hole.
Scientific claim N-B — the *k*-scaling law	**ACHIEVED ON ONE DATASET**	Complete on VitalDB: 24/24 configurations, monotone and significant in AUPRC, with a built-in positive control at $k = 1$. **Not yet shown to generalise.** AUROC is directionally consistent but underpowered at n = 3. Establishing whether this is a structural law or a dataset-specific artefact is the single highest-value remaining experiment.
Scientific claim N-A′ — head-only DP memorises	**ACHIEVED**	0.6277 under LiRA versus 0.5122 for full-model training, $p = 1\times10^{-9}$ over 6 runs. A real caveat to the parameter-efficient DP fine-tuning literature, and self-critical since it concerns this project's own chosen recipe.
Claim N-1 — genuinely paired multimodal benchmark	**ACHIEVED, WITH LIMITS**	+0.0370 AUROC at 5 seeds on VitalDB, 5.9× seed sd. But WEAR is at ceiling and cannot corroborate it, so the claim rests on **one dataset**. Reported alongside the finding that some IoT tasks do not need fusion at all.
Claim N-D — normalised patient clipping	**NOT STARTED**	Identified but untested. The known flaw: a patient's gradient is summed over *k* windows and then clipped to a fixed bound *C*, so long recordings are systematically down-weighted. The proposed fix normalises by *k* or $\sqrt{k}$ before clipping. This is the **only remaining candidate for an original mechanism** rather than a measurement.

PART VI — COMPLETE FINDINGS REGISTER

ID	Claim	Strength /10	State	Evidence
N-B	Cost of the guarantee scales with records per patient	8	**MEASURED**	24/24 configs; gap 0 → +0.0507 → +0.0832 → +0.0866 AUPRC; control at k=1 (p=0.75)
N-D	Normalised patient clipping beats fixed-C	8	NOT RUN	—
N-A′	Head-only DP fine-tuning memorises members	7	**MEASURED**	0.6277 LiRA vs 0.5122 full-model; p = 1e−9, sd 0.0018 over 6 runs
N-1	Genuinely paired multimodal benchmark shows fusion gain	7.5	**ONE DATASET**	+0.0370 AUROC / +0.0671 AUPRC, 5 seeds, 5.9× sd (VitalDB); WEAR at ceiling
N-2	Patient-level DP cost measured	7.5	**DONE**	Superseded in importance by N-B, which generalises it
N-3	Cost of a *valid* guarantee (disjoint pretraining)	6.5	**MEASURED**	−0.022 AUROC versus oracle pretraining
N-4	Head-only DP viable in hierarchical multimodal FL	6	**MEASURED**	Gap widens as budget tightens; holds on both datasets
N-5	Adaptive-DP composition tax	5.5	**NEGATIVE**	Cross-group split drives σ 11.3 → 70; worse than uniform at equal ε
—	DP shifts model attention toward missingness	5	**MEASURED**	Mask attribution share: full_dp 28.2 % > patient 22.0 % ≈ sample 21.6 % > full_nodp 19.9 % > head_nodp 15.8 %; rank ρ = 0.95
N-6	Specialty-derived non-IID partitioning	4	**MEASURED**	3.2–50.7 % divergence across clients
N-7	Missingness carries signal	4	**EVIDENCE, NOT PROOF**	Masks carry 22.0 % of attribution mass over 5 seeds; `ART_MBP_observed` 4.99 % ± 0.33, top-5 on 4/5 seeds. Say “consistently among the strongest”, not “#2”. Needs the mask ablation to become proof
~~N-A~~	~~Patient MIA: record-level DP is exploitable~~	—	**REFUTED**	All four attack Δ intervals include zero; control passed at 0.6277. Kept as a validated negative

*Table 8 — Dropped entirely: CMGA, FedKD-E, pFAL, ASWA, the PHANTOM-FL brand, and all intellectual-property framing. **No claim in this project is patentable**, and framing any of it as IP would invite rejection.*

PART VII — OPEN ISSUES AND NEXT STEPS

27. THE IMMEDIATE BLOCKER

External clinical validation data (eICU-CRD) is not yet accessible. A request for PhysioNet credentialing is in progress; the guide has been asked by email to act as reference. Until that clears, the experiment that would establish whether the k-scaling law is structural or dataset-specific cannot run.

> **THIS DELAY IS USABLE, NOT MERELY LOST TIME**
>
> The scientific value of the eICU experiment depends on the prediction being **frozen before the data can be seen**. If the curve is fitted, eICU's k distribution inspected, and the prediction written afterwards, no reviewer can distinguish that from retrofitting — and neither can we. The waiting period is therefore exactly when the pre-registration must be written. Nothing else about the restructuring is blocked on access.

27.1 What is blocked, and what is not

Work item	Blocked on data access?
Freeze the k-law prediction with intervals	**Must be done BEFORE access**
Promote the k-law to lead contribution in the manuscript	No
Restructure Results around the law	No
Redraw Fig. 4 from real LiRA scores (currently synthetic)	No
Demote fusion to a setting result with 5-seed numbers	No
Build the eICU pipeline against the published schema	No
Run the eICU arms	**Yes** — the only genuinely blocked item

28. OTHER OPEN ISSUES

28.1 Rigorous public pretraining

AFFECTS THE ε CLAIM — the encoders in several runs saw private data non-privately. Closing this needs either an external public corpus (PTB-XL) or a disjoint public index split. The proxy-versus-public gap is already priced at −0.022 AUROC, so the claim is quantified even where it is not yet closed.

28.2 Figure 4 is illustrative, not evidential

The LiRA figure currently uses synthetic curves and is captioned "Illustrative". The measured per-patient IN/OUT distributions now exist in the shadow-model score files. Redrawing from those converts a teaching diagram into evidence — a cheap, high-value fix.

28.3 Statistical power

The *k*-scaling law rests on n = 3 seeds. AUPRC resolves; AUROC does not. Raising the seed count would let both metrics be reported as resolved rather than one as "directionally consistent".

28.4 Ablations A1–A4

Fixed on 16 August 2026 after four silent bugs, but the arms still need to be run to populate the component ablation table.

28.5 Claim N-D remains untested

Normalised patient clipping is the only remaining candidate for an *original mechanism* rather than a measurement of an existing one. If it wins it materially raises the contribution; if it loses it costs little. It is low-effort and should be run.

29. THE RECOMMENDED PATH FORWARD

1. **Pre-register the *k*-law prediction** — fit the curve on VitalDB, freeze a numeric forecast with intervals, commit it dated. Must precede data access.
2. **Restructure the manuscript** so the *k*-law is the first contribution rather than the fifth, with hierarchy, compression, energy and latency as the vehicle that makes the finding deployable rather than as the headline.
3. **Write the eICU pipeline blind** against the published schema, so that nothing waits on credentialing.
4. **Run the comparison** when access lands, and report the outcome either way.
5. **Run claim N-D** in parallel, since it is independent of all of the above.

30. WHY EITHER EICU OUTCOME IS PUBLISHABLE

If the prediction lands, the cost of a patient-level guarantee is shown to be **structural rather than dataset-specific**, and the work delivers a transferable design rule: *given your k, here is what patient-level ε = 1 will cost you*. That is what a practitioner actually needs, and it has not been published.

If the prediction misses, the work has found a **boundary condition**, with a registered forecast proving the story was not retrofitted. That is also publishable, and visibly honest.

A paper reporting that two independent clinical datasets were tested, one
reproducing the result and the other exposing a limitation, is more
publishable than one in which everything conveniently succeeded.

---

31. CLOSING ASSESSMENT

The engineering objectives are met with wide margins and are not in doubt. The project's credibility rests on a pattern it has followed consistently: pre-registering gates before running attacks, reporting the inconclusive WEAR result rather than extending seeds until it passed, withdrawing an entire dataset when its pairing proved invalid, correcting a published effect size downward by 37 % after replication, and deleting components that measured as contributing nothing.

That discipline is why a refuted central hypothesis did not sink the work. It is also the strongest argument that the *k*-scaling law — which arrived with its own positive control built into the experimental design — means what it appears to mean.